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Demand characteristics in human–computer experiments

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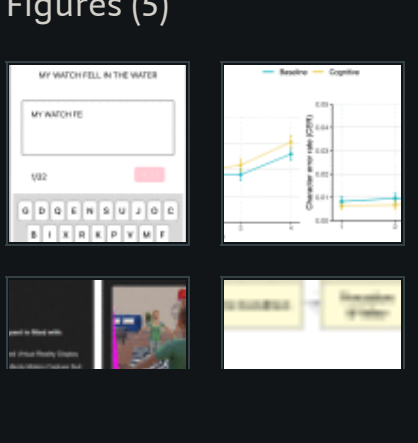

Highlights

- Demand characteristics are cues that can inform participants about the hypothesis.
- Demand characteristics can influence participant's behavior.
- Demand characteristics often go uncontrolled, especially for objective measures.
- Demand characteristics affect experiment outcomes even when experimenter is absent.
- Demand characteristics may confound both objective and subjective measurements.

Abstract

Demand characteristics refer to cues that can inform participants in experiments about the hypothesis and influence their behavior. They lead researchers to erroneously infer non-existing effects, undermining the experimental integrity of empirical studies. Despite a widespread acknowledgment of their confounding influence in experimental psychology, experiments involving humans and computers to a lesser extent consider effects of demand characteristics, as computerized protocols are thought to be immune to some experimenter biases. Furthermore, demand characteristics are considered to mainly effect subjective measures. As a result, demand characteristics often remain uncontrolled in studies involving computers, and in particular for objective measures such as performance.

In this paper, we present two experiments that underline the importance of demand characteristics in human–computer interaction experiments. In a text-entry study, we made participants believe they were evaluating a research-based keyboard. This belief led to increased performance and self-reported user experience. In a second study, we conducted a thought experiment on the illusion of body ownership in virtual reality, where the experimental design indicated the study hypothesis. We found hypothesis-compliant responses from participants, even when they did not experience the illusion. We conclude that demand characteristics pose a significant challenge to the interpretation and validity of human–computer experiments, even when they are fully automated. We discuss the implications and offer guidelines to mitigate effects of demand characteristics.

Keywords

Demand characteristics; Experiments; Confounds; Bias; Validity; Human–computer interaction

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1. Introduction

Participants in scientific experiments naturally form beliefs about a study's hypothesis and purpose by observing various cues in the experiment situation. If participants modify their behavior in line with their beliefs, it can influence the study's outcomes. Orne (1959) drew attention to this phenomenon, coining the term "demand characteristics".

Demand characteristics have turned out to be a crucial external variable influencing psychological experiments. With mounting evidence of their subtle confounding effects across the spectrum of experimental psychology, nagging questions have arisen about the validity of much highly influential experimental research (Elms, 1975). Orne's later critical examination drew attention to possible cues in some of the most famous and foundational experiments in social psychology, such as Zimbardo's prison experiment (Haney et al., 1973) and Milgram's work on obedience (Orne, 1962). Orne argued that the findings from these studies might be attributable to participants merely playing roles they believed were expected of them rather than reflections of other causal mechanisms.

The specter of demand characteristics looms large beyond social psychology, casting a shadow over the validity of foundational studies in other fields of behavioral science. Recent work by Lush (2020) drew attention to the potential impact of suggestion effects in the rubber hand illusion. More recently, a comprehensive meta-review by Coles and Frank (2023) broadened the perspective on how demand characteristics can infect research results by showing that the issue can rear its head in any non-clinical experimental study in psychology. When their effects are uncontrolled, demand characteristics can systematically lead to hypothesis-compliant responses. This growing body of evidence has prompted discussion of an apparent crisis of confidence in psychological studies (Sharpe and Whelton, 2016).

Neighboring disciplines are not immune to these effects. Human–computer interaction (HCI) scholarship draws especially heavily on the traditions and practices of the behavioral sciences and experimental psychology. Adjacent disciplines typically give demand characteristics considerable attention (Lush, 2020, Corneille and Lush, 2023, Coles and Frank, 2023), yet the HCI discipline seems to have engaged less with such experimental biases — even though its empirical work likewise relies on human participants. A reason for this lack of engagement with the concept may be the common perception that demand characteristics are irrelevant to computerized settings. For example, scholars, such as Donohue et al. (1999) asserting the non-generalization of demand characteristics to computers, posit that increased perceived anonymity fosters participants' inclination to respond honestly and naturally (Epstein et al., 2001, Naus et al., 2009, Bates and Cox, 2008, Sawhney and Cigularov, 2014). The allure of web-based research, as explored by Ryan et al. (2013), resides in the advantages of large participant pools and the absence of traditional demand characteristics associated with experimenter guidance (Ryan et al., 2013, Yu and Wang, 2015, Casler et al., 2013, Epstein and Klinkenberg, 2001). Yet, even as these studies leverage fully automatized computerized protocols, the inherent human consciousness and psychology of participants persist, shaping their interpretation of the experimental situation and its purpose.

The assumption that demand characteristics predominantly pertain to subjective reports is also a prevailing notion. Scholars often regard objective measures, such as behavioral or physiological measures, as immune to these influences, using them as benchmarks for validating subjective reports (Vraga et al., 2019, Muter et al., 1993, Lin et al., 2023, Read et al., 2016).

Existing literature delving into the influence of demand characteristics in the realm of human–computer interaction reveals a discernible impact on subjective evaluations of artifacts. Notably, participants exhibit a proclivity for the artifact endorsed by the researcher, as observed in studies such as those by Dell et al. (2012). In this interview study, when the researchers hinted that an artifact was developed by them, participants were more than twice as likely to express a preference for that artifact, even when the alternative was identical or even superior. A parallel exploration of the placebo effects, as documented by Kosch et al. (2023) and Denisova and Cairns (2015), further underscores the intricate interplay of demand characteristics with participants' reported experiences, particularly when they believe they are engaging with artificial intelligence (AI). These investigations shed light on the pivotal role of demand characteristics in experimental laboratory research that involves the intersection of human participants and computers. However, in HCI, their influence seems to be considered only in situations involving subjective preferences and in the presence of human evaluators. The broader implications of these cues beyond such settings remain shrouded in uncertainty. While research in diverse fields hints at connections with an array of methodological concerns (McCambridge et al., 2012, Weber and Cook, 1972), the extent of the validity challenges they pose for computerized experiments remains elusive.

To bring the influence of demand characteristics on empirical research in HCI into the open, we performed two experiments. A text-entry study broadened the scope beyond subjective reports by examining objective metrics for typing speed alongside self-reported user-experience (UX) measurements. This revealed that demand characteristics can indeed confound objective performance metrics, even without the presence of a human evaluator. With the second experiment, which mirrors studies of body-ownership illusions, we ventured into a more immersive domain of HCI, virtual reality (VR), in which the interaction with technology is indirect. Experiments in VR are highly controlled and considered to be devoid from contextual confounds (Pan and Hamilton, 2018, Wilson and Soranzo, 2015, Vraga et al., 2019). We demonstrated how a commonplace choice of experimental design and procedure may help participants guess the hypothesis, thereby influencing their responses in the conditions presented, even when the study is fully automated. These two studies show that demand characteristics in experiments involving computer use are crucial, expanding upon the mainstream notion that they interact merely with preferences (Dell et al., 2012). Demand characteristics, as such, pose a danger in human–computer interaction experiments, also with validated self-reporting instruments, and even in work considering objective performance metrics.

Our contribution is twofold. The paper documents the confounding influence of demand characteristics on both objective and subjective measurements in settings with several key similarities to the bulk of HCI experimental research. Second, we go further by outlining some recommendations in light of our results. While we acknowledge that completely eliminating demand characteristics is impossible, scholars can diminish these insidious cues' influence on their work.

2. Background

2.1. Demand characteristics

When Orne (1959) introduced the concept of demand characteristics , he defined them as a "totality of cues that convey an experimental hypothesis to the subject" and "become significant determinants of subjects' behavior", thereby posing "a serious threat to the validity of the study results". His experiments demonstrated that participants often strive to provide what they believe are desired results, even if the tasks are tedious or seemingly pointless. This revelation called into question the assumption that experiments can isolate the direct causal relationship between independent and dependent variables. Human subjects, being conscious, inherently try to make sense of what is happening to them. It leads to guesses about the purpose of the experiment, which becomes an uncontrolled confounding variable and makes demand characteristics a significant confounding factor in experimental settings (Orne, 1962).

This insight prompted a reevaluation of numerous influential studies in social psychology, of phenomena such as classical conditioning (Page, 1969), situated social cognition (Smith and Semin, 2004), and priming (Doyen et al., 2012). Orne raised doubts about the authenticity of participants' behavior in general, contributing to the broader scholarly discourse that they might merely be playing roles to conform to perceived expectations (Orne and Holland, 1968, Banuazizi and Movahedi, 1975). Although participants may often adjust their behavior unintentionally, demand characteristics can also lead to deliberate changes in behavior and influence subjective experiences, both of which can contaminate experimental results (Orne and Scheibe, 1964). As doubts swept the field, efforts to replicate such accepted phenomena as the "weapon effect" (wherein the presence of weapons seems to trigger aggressive behavior) failed when the hypothesis was concealed from participants (Berkowitz and LePage, 1967, Page and Scheidt, 1971).

In the wake of Orne's groundbreaking work, experimental psychologists recognized the need to address demand characteristics in their research. They began designing experiments in ways that conceal the true objectives from participants. With these and other manipulations, this constellation of cues became a fundamental component of the field's research methodology (e.g., Shaughnessy et al., 2000). The aim was to reduce the influence of participants' inclination to align their behavior with the hypotheses as perceived.

Although demand characteristics are currently receiving attention in experimental psychology, this has not always been the case. The interest of Orne's work in the 1960s (Orne, 1962, Orne and Scheibe, 1964, Orne and Holland, 1968) declined from the 1980s and onward (Sharpe and Whelton, 2016, Klein et al., 2012). Psychology scholars only recently reignited their focus on the issues surrounding demand characteristics in research (Sharpe and Whelton, 2016, Klein et al., 2012).

Recent insights suggest that the effects of demand characteristics may indeed be more intricate, potentially confounding empirical research in subtle ways. In an attempt at showing the existence of an effect due to an intervention, researchers may choose to add a control condition wherein the effect is expected to be diminished. Such experimental designs may unintentionally reveal the study's hypothesis, even if it is not communicated by an experimenter. In this vein, Lush et al. (2020) suggested that the Rubber Hand Illusion (RHI) (Botvinick and Cohen, 1998) might be partly due to suggestion effects. In RHI, participants experience ownership of a fake limb after synchronous stimulus of the fake and real limb. The critique suggests, that when participants are exposed to the rubber hand illusion, they become aware of the researcher's expectations, which might activate phenomenological control. Phenomenological control refers to the capacity to alter subjective experience such that it misrepresents reality in ways consistent with experimenter's goals. Hence, participants can experience the RHI due to demand characteristics effects. Demand characteristics may as such confound the validity of studies in subtle ways that can be hard to mitigate.

2.2. Demand characteristics in HCI studies

Examination of demand characteristics in HCI experiments is rare, especially relative to the volume of experiment-based work in this discipline. Infrequent studies that account for demand characteristics, mostly discuss the possibility of the observed results being affected by demand characteristics effects (e.g., Duggan et al., 2013, Kätsyri et al., 2017, Hoorn and Huang, 2024, Duggan et al., 2012, Lee, 2003, Morgan et al., 2022, de Kleijn et al., 2019, Geeseman and Balters, 2022), though rarely do they take any measures to control for them or mitigate their effects, with only a few notable exceptions (e.g., Cairns et al., 2013, Johnson et al., 2004, Yarosh et al., 2022, Stevens et al., 2013). The common method to avoid demand characteristics in the literature is not to provide any information about the study (Teo et al., 2003, Shin et al., 2019). However, the concept of demand characteristics suggests that the study hypothesis can be discerned just from the experiment's situation and the way the experimental procedure is set, potentially influencing participants' behaviors in line with expected outcomes, and thereby compromising the internal validity (Orne, 1962). Although there are some prominent positive examples of measures taken to control for the influence of demand characteristics, the proportion of studies that take them into account is intolerable compared to the amount of experimental research taking place in the field of human–computer interaction.

Below, we provide an overview of the contributions that have addressed them in the context of HCI research.

2.2.1. User studies

The problem of demand characteristics undermines techniques employed in experimental HCI work. Subjective self-reporting is the most obvious issue. Many HCI experiments involve subjective evaluation of artifacts, and these cues loom large in that context. Accordingly, Dell et al. (2012) conducted two experiments asking participants to indicate which of two mobile video-players they preferred. One was introduced as designed by the interviewer. Even when it produced inferior video quality, the player presented as the interviewer's creation was preferred. Although the paper highlighted the biased evaluations well, it neither probed the vast range of response types that can be influenced nor considered the wider spectrum of possible cues. Furthermore, the HCI community has not acted on the paper's conclusions even within the area highlighted, and only a few articles have mentioned demand characteristics in connection with designing or interpreting user studies (Jansen and Hornbæk, 2018, Phelan et al., 2019), with one notable exception being Lyngs et al. (2020) controlling for demand characteristics.

We can turn to research on demand characteristics in psychology for what it reveals about phenomena central to the HCI field. One example is Lush's aforementioned work showing that the rubber hand illusion may stem in part, if not entirely, from demand characteristics. This illusion has formed the basis for HCI research on avatars in virtual reality (VR). Related work has found also that the experience of embodiment and presence in VR is moderated by participants' understanding of the behavior expected from them (Forster et al., 2022).

It bears reiterating that experimental HCI research extends far beyond subjective self-reporting. While work pertaining to the former alone offers evidence that HCI research could increase its validity through adequate attention to demand characteristics, we need to go further. Again, their relevance for other ways of collecting data experimentally – with validated scales, objective measurements, and immersive experiences – has not yet been explored.

2.2.2. Studies in the wild

Studies "in the wild" are generally considered to offer advantages that controlled settings do not, such as capturing unanticipated uses of technologies. They are less artificial and lack strictly specified hypotheses, but demand characteristics still rear their head here. Brown et al. (2011) demonstrated this with regard to a photo app. Users adjusted their behavior in the wild to match the researcher's expectations: they took more photos than usual, and they coordinated their use of the system with each other so as to provide more data about interactions involving the app.

2.2.3. The placebo effect

In medicine, a sham treatment (for example, a water injection) can improve the patient's condition, by subjective criteria (Hróbjartsson and Gøtzsche, 2001, Finniss et al., 2010). This is the well-known *placebo* effect. Human–computer interaction researchers have recently engaged with this concept in the domain of artificial intelligence and cognitive augmentation.

Just as patients experience a positive placebo effect on their health from inactive pills only if having a prior belief that pills are good for one's health, beliefs about AI constitute a major focus in current HCI. There is a popular conception that this technology can markedly enhance the user experience. Hence, Denisova and Cairns (2015) found that the gaming experience was perceived as better when the player believed AI to be aiding with skill-level-based complexity adjustments, even though no AI was acting. Similarly, Kosch et al. (2023) found that priming participants for an AI-augmented puzzle game changed performance expectations and affected usability metrics. People form similar beliefs about cognitive assistance. For instance, after being told of the positive influence of a sound developed by cognitive researchers, participants showed an increase in risk-taking behavior in a cognition-based task (Villa et al., 2023). These findings serve as important evidence that HCI domains can suffer from suggestibility related to a study's expected or desired outcomes.

2.3. Summary

The effect of demand characteristics has been demonstrated in multitudes of contexts, from user experience with novel technologies to cognitive illusions. Yet, several questions about their manifestations in human–computer interaction have gone unanswered, because HCI-specific studies of demand characteristics have addressed only their impact on results from experiments that rely on self-reported data. Therefore, whether demand characteristics affect performance values has been open to question. If they do, we must revisit the validity of past and future HCI experiments employing measurements generally regarded as unbiased indicators of real behavior. The earlier works have also employed quite overt manipulations to identify the effects of demand characteristics on experimental outcomes (Dell et al., 2012), while not all studies communicate their research hypothesis as explicitly. This raises the question of whether demand characteristics exert an influence when the hypothesis is presented in a more subtle manner.

3. Experiment I: Interface evaluation

Experiment I was designed to assess how demand characteristics confound measurements, in absence of an evaluator, in a fully computerized study. The experiment presents a typical HCI study in which participants directly interact with technology. We compared participants performance in entering text using two interfaces. Text-entry represents a common computer activity, and text-entry studies are a prominent class of empirical research in human–computer interaction. Text-entry studies also represent research on apprehension of novel technology, as such studies typically compare a research-based user interface with a baseline (e.g., Fukumoto and Tonomura, 1997, Soukoreff and MacKenzie, 2003, James and Reischel, 2001, MacKenzie and Soukoreff, 2003, MacKenzie, 2007, Zhai and Kristensson, 2003, Oney et al., 2013, Markussen et al., 2014, Mottelson et al., 2016). In typical user interface evaluations, the obviousness of the experiment's intervention can render it challenging to isolate the effect of the interface. For example, the baseline condition is often an interface that participants are already familiar with. A text-entry study offered an advantage in this regard: with a keyboard we were able to develop two prototypes that are functionally identical but look different. We could isolate and test the effects of demand characteristics without confounding them with real differences between the artifacts. Finally, decades' worth of attention to text-entry has yielded standardized objective metrics. Therefore, this activity is ideal for examining the influence of demand characteristics, both subjectively and objectively, employing an experimental design that emulates authentic research frameworks.

Reviewing text-entry studies

To understand how text entry studies are typically conducted, we queried the ACM Digital Library for text entry studies empirically comparing two or more text entry interfaces. In reviewing these 137 text-entry studies we found that within-subjects designs (112) are more than four times as prevalent as between subjects design (25). Text-entry studies commonly have one independent variable (keyboard) comparing a baseline and research-keyboard (all studies); circa half of the studies compared three keyboards, 15% compared four or more keyboard variants. Extracting typing speed means and standard deviations for all baseline and research keyboards allows us to compute the average effect size. We find that the experimental keyboard outperforms the baseline considerably (Cohen's $d = 1.88$). Data and code for the text-entry review are available at OSF.[1] We use these observations in designing a study that follows common experimental design choices, as to underline demand characteristics presence in research following common practices in interface evaluation.

3.1. Hypotheses and preregistration

We presented two software keyboards to the participants, identifying one as fruit of the research team's development efforts based on findings from cognitive science (Cognitive) and the other as a baseline keyboard (Baseline). The keyboards were in fact functionally equivalent. The study aimed to demonstrate that participants would adjust their behavior to their conceptions of the study's aim. We hypothesized that performance would be better with the keyboard connected with the description in the Cognitive condition, compared to the Baseline condition. Specifically, we hypothesized that the research-based grounding for the keyboard layout in the Cognitive condition would encourage higher text-entry speeds, lower error rates, and stronger preference. We also outlined an exploratory analysis about the possibility of UX ratings being higher in the Cognitive condition.

The research received approval from the host institution's ethics review board and the Apple App Store review team in fulfillment of their guidelines on human-subject research. The study, its hypotheses, and the analysis approach were preregistered with AsPredicted.[2] The data and analysis code have been archived at OSF.[3]

3.2. Participants

We recruited 120 native English-speakers to take part (60 male, 59 female, 1 non-binary) via the crowdsourcing platform Prolific. The number of participants was selected as a trade-off between statistical power and resource constraints. Notably, this sample size substantially exceeds the mean of 22 participants commonly observed in classical text entry studies, ensuring robustness in our findings. The ages ranged between 18 and 55 years ($M = 32.3$, $SD = 9.4$). About 88% (106 participants) were predominantly right-handed. The experiment took approximately 50 min, and was compensated £10 . Following the protocol specified in the preregistration stage, we excluded five participants whose mean character-error rate (CER) exceeded 0.05. This left 115 participants' data for the main analysis.

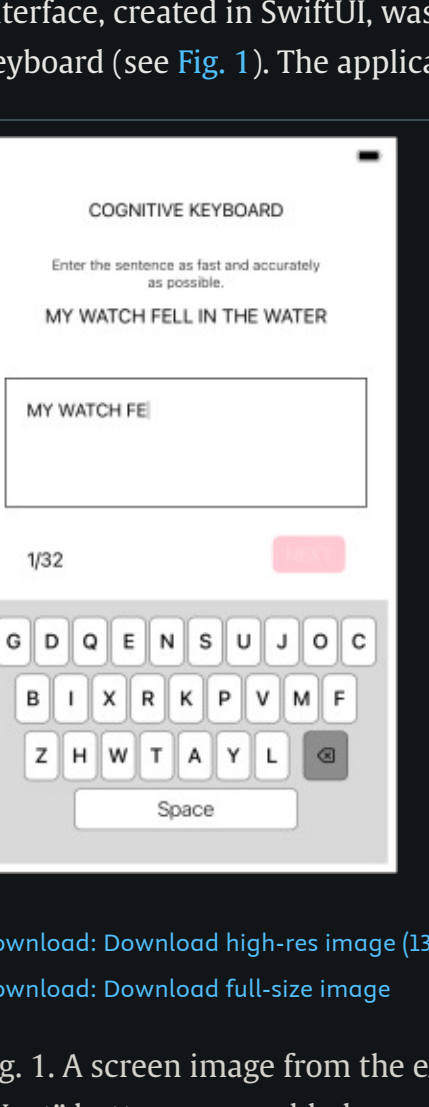
3.3. Apparatus

The mobile application

We developed an iPhone application for participants to install on their own devices. The application, distributed via Apple's App Store, targeted iOS 13 and above (i.e., model 11 and later iPhones), to guarantee comparable device layouts. Upon completion of the experiment, the application sent relevant user metrics over HTTPS to a server application written in Python, hosted by Google App Engine. The application enforced full-screen portrait orientation.

Keyboard layout

The keyboard layouts were generated uniquely for each participant, via the same algorithm in every case. By generating unique keyboard layouts for each participant, we guarded against unintentionally using a superior layout for one condition across all participants. The logic for letter positions began with random placement of a random letter, then maximized the distance between high-frequency letters. The algorithm thus guaranteed considerably different yet functionally equivalent layouts given different seed letters. Our aim with this design was to manipulate only the explanation of the keyboards. The keyboard interface, created in SwiftUI, was designed for similarity to the standard iOS keyboard (see Fig. 1). The application enforced a bright layout.


COGNITIVE KEYBOARD

Enter the sentence as fast and accurately as possible.

MY WATCH FELL IN THE WATER

MY WATCH FE

1/32

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Fig. 1. A screen image from the experiment's application during use. The pink "Next" button was enabled once the number of spaces in the target sentence and the user's transcription matched.

3.4. Design

The experiment was designed to follow established norms in text-entry studies, following state of the art (MacKenzie, 2007) of text entry research. The study hence followed a random-ordered within-subject design addressing one independent variable, keyboard, with two levels: Baseline and Cognitive. Both conditions featured soft keyboards that measured participants' text-entry performance. The two levels employed functionally equivalent keyboard layouts, as described above, with different explanations of their origin, shown in Table 1.

The baseline keyboard.

For the Baseline condition, we explained that the keys in the layout were randomly shuffled, with subsequent ordering to distribute the letters on the basis of their frequency in English text.

The cognitive keyboard.

For the Cognitive condition, participants were told that the layout had been designed by means of heuristics from cognitive research that take advantage of memory capabilities in pursuit of faster typing. It was kept secret from participants that the cognitive keyboard was functionally equivalent to the baseline layout (i.e., that the same algorithm had generated both).

3.5. Dependent variables

Speed.

We assessed typing speed in words per minute (WPM) (Yamada, 1980), using the `wpm` Python package,[4] which calculates WPM by dividing the number of characters typed per minute by 5.

Character-error rate.

Participants' error rates were calculated as "edit distance", the minimal change required, in characters, for correspondence between the target text and that entered (MacKenzie and Soukoreff, 2002), by means of the `jiwer` Python package.[5]

User experience.

To measure subjective user experience, we employed 10 items from the System Usability Scale (SUS), each with a five-point Likert scale (Brooke, 1996). The final score is calculated by subtracting 1 from the score for the odd-numbered questions and subtracting each value for the even-numbered questions from 5, then summing the resulting numbers and multiplying the sum by 2.5 (Brooke, 1996).

Preference.

Finally, participants were asked which of the keyboards they preferred, in a forced binary choice.

3.6. Procedure

Once they had given informed consent and provided demographic information, participants began the experiment. They were introduced to one of the two keyboard layouts, selected randomly, then were requested to transcribe sentences as quickly and accurately as possible (after entering five sentences as warm-up).

The study was divided into four eight-trial blocks, with a 30-second break between blocks to reduce fatigue, as is established practice for text-entry studies (MacKenzie, 2007). Accordingly, each participant completed 8 trials $\times$ 4 blocks $\times$ 2 layouts $= 64$ sentences. With 120 participants, the dataset comprised 7680 completed trials. Each trial screen featured a header showing which keyboard the participant was currently testing, instructions, the sentence for transcription, an area for entering the text, and a progress indicator. Pressing a "Next" button allowed the participant to advance to the next sentence. Fig. 1 presents the interface.

Table 1. The difference between the two conditions in Experiment I lay only in the instructions and explanations for the keyboards used. The table shows the verbatim text provided before trials commenced, for both conditions.

	Baseline	Cognitive
Instruction	Now, you will test the baseline keyboard.	Now, you will test a research-based keyboard that incorporates advanced cognitive heuristics.
Explanation	This keyboard is randomly shuffled. However, we implemented a frequency-based heuristic such that commonly used letters are not placed next to each other. This makes the keyboard similar to the standard layout.	We implemented mnemonic associations by arranging keys in patterns that resemble their visual appearance. This can help users quickly learn and remember the key positions. We also implemented chunking, by arranging frequently used letter combinations and sequences together. This allows recalling and typing common patterns more easily and quickly, leveraging memory capabilities. These heuristics are intended to optimize typing speed and accuracy.

3.7. Results

Speed.

The mean WPM figure favored the Cognitive keyboard (see Fig. 2, left): $M = 11.17$, $SD = 3.63$, while Baseline yielded $M = 10.86$, $SD = 3.57$, with Cohen's $d = 0.30$. Using repeated-measures ANOVA to compare text-entry speeds between the two keyboards, we found a significant main effect of keyboard: $F(1, 114) = 6.90$, $p = 0.009$. We also found a significant main effect of block, illustrating the learning effect for both keyboards (

$F(3, 342) = 79.58, \quad p < 0.001$). No interaction effect between keyboard and block was visible ($p = 0.25$).

Character-error rate.

Participants made few errors and showed comparable error rates across conditions (see Fig. 2, right) – Cognitive: $M = 0.007, \quad SD = 0.026$; Baseline: $M = 0.008$, $SD = 0.029$. Comparison using repeated-measures ANOVA did not reveal a significant effect for keyboard: $F(1, 114) = 2.27, \quad p = 0.14$; or block: $F(3, 342) = 0.53, \quad p = 0.64$.

User experience.

The mean reported UX score was higher for the Cognitive keyboard ($M = 46.9, \quad SD = 22.4$) than for the Baseline one ($M = 42.0, \quad SD = 18.3$). A t-test revealed a significant difference: $t(114) = 2.8, \quad p = 0.006$, Cohen's $d = 0.24$.

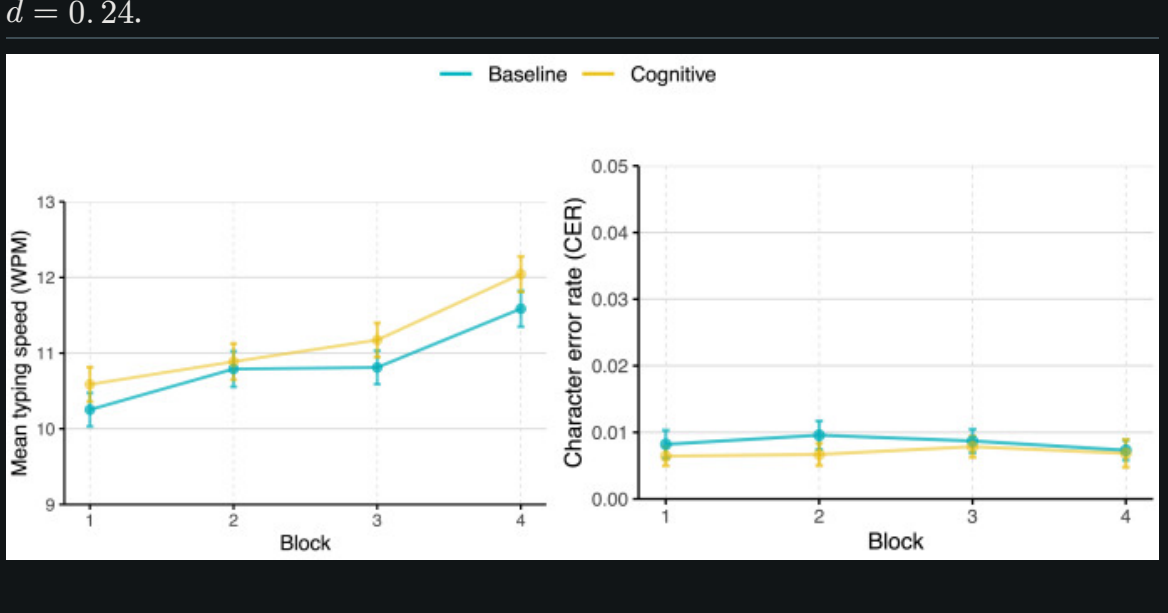

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Fig. 2. Block-to-block development of speed (a) and error rates (b) with the two keyboards. For the one portrayed as designed with heuristics, text-entry speeds were significantly higher. Error rates showed no effect. Error bars denote 95% confidence intervals.

Preference.

Participants expressed a preference for the Cognitive keyboard and the Baseline one in roughly equal measure (53% and 47%, respectively) in the forced binary choice.

3.8. The overall findings

The study revealed a significantly higher typing speed for the "research-based" Cognitive keyboard, relative to a functionally identical layout that yielded similar error rates. This was accompanied by significant differences in perceived user experience. Together, these results confirm our hypothesis that performance and experience results in user studies can be affected by the implication that one is taking part in an experiment aimed at confirming the effectiveness of a novel interface. Compared to the effect sizes reported in the literature (Cohen's $d = 1.88$), our effect sizes for text-entry speeds are small (Cohen's $d = 0.30$), which, given that the keyboards were functionally identical, illustrates that demand characteristics could potentially amplify even modest effects enough to surpass the significance threshold. These results offer evidence of the influence of demand characteristics in a typical HCI study utilizing established empirical protocols. Without traditional evaluator oversight, it was observed that participants exhibited significantly higher subjective scores and objective values of performance when interacting with an interface that they believed to have been developed by the researchers.

It is important to recognize that our understanding of the observed effects is constrained by the lack of measurement of participants' expectations, such as their perceived improvements in speed when using the Cognitive keyboard. If these expectations only slightly alter, the effects on dependent variables may not solely be due to demand characteristics but could also stem from placebo effects (Boot et al., 2013). This development shifts the focus from differentiating between them to emphasizing the critical need for their meticulous management in research studies.

4. Experiment II: Immersive experience

In the second experiment, we aimed to reveal the existence of demand characteristics effects in a field of HCI concerning experiential computer use. We selected a class of experiments that concern body illusions using virtual reality (VR) technology. The studies in VR are considered to be highly controlled and immune from the confounding influence of contextual variables (Parsons, 2015, Pan and Hamilton, 2018, Wilson and Soranzo, 2015). The study examined how participants may guess the hypothesis from the experiment design and alter their responses accordingly. This endeavor aims to unravel the subtle yet influential role demand characteristics play in shaping participant responses within the context of immersive experiences.

The chosen experimental paradigm revolves around a well-established body-ownership illusion in VR. Participants undergo an experience wherein their real body is seemingly replaced by a virtual body, synchronized with their actual movements. Body-ownership illusions have long captivated cognitive science and psychology, and have been claimed to provide a rich foundation for studying the intricate relationship between the body and the mind (Maister et al., 2015). This specific illusion is meticulously designed, featuring a synchronous condition where the virtual body mirrors real-world movements, which through multisensory integration makes people experience ownership of the virtual body (Kilteni et al., 2015). The standardized experiments on body ownership illusions typically include an asynchronous control condition wherein the virtual body moves independently of the participant.

These illusions offer a fundamental experiment-based paradigm for enhancing user experiences in VR and for studying how the body shapes the mind. Any effect observed in the synchronous condition, implemented by tracking a person's real-world body and rendering a synchronously moving virtual one (an avatar) in real time, is considered to indicate that the measures reflect multisensory integration. This inference, is drawn only if the same effect is absent under the asynchronous condition (e.g., Osimo et al., 2015, Jung et al., 2020, Kondo et al., 2020). Researchers traditionally employ asynchrony as a control condition to compare against the treatment condition and thereby identify causal links between manipulations and dependent variables (Banakou et al., 2013, Banakou et al., 2016, Tajadura-Jiménez et al., 2017), such as between virtual skin color and racial bias (Peck et al., 2013).

In the landscape of experimental psychology and HCI, studies on body ownership often adhere to standardized protocols, incorporating the asynchronous condition as a crucial element to validate experimental manipulations and a recent review points to more than 100 of such VR experiments from the last decade alone (Mottelson et al., 2023). Notably, it highlights a significant distinction in effects, particularly concerning the within-factor of visuo-motor synchrony. Within-subjects designs were found to exhibit a notably larger effect in embodiment scores (Hedges $g = 0.97$), approximately twice the size compared to between-subjects designs (Hedges $g = 0.45$), and 74% of studies on body ownership in VR employing asynchronous and synchronous conditions utilize within-subjects designs. This difference is attributed to the frequent utilization of visuo-motor asynchrony as a within-subjects condition, which makes the expected difference in body ownership scores obvious to participants, emphasizing it as the most effective manipulation for body ownership illusions (Mottelson et al., 2023). It is noteworthy that when comparing only manipulations of visuo-motor synchrony, there is no significant difference in body ownership effect sizes across experimental designs. This nuanced observation underscores the critical significance of within-subjects designs, particularly in the context of investigating demand characteristics within the domain of visuo-motor synchrony. An additional danger lies in the fact that experimental design can determine participant's experience of the illusion.

Inspired by Lush (2020) and Corneille and Béna (2023), we employed the method of instruction-based replication. This approach is designed to determine if experimental outcomes are influenced more by participants' expectations and common sense knowledge than by the actual experimental manipulations. Under this method, participants are briefed about various aspects of the study, including the experimental procedure, tasks, and stimuli. Then, they are asked to reflect on being exposed to the stimuli described and consider what their response would be had they participated in the experiment. If their guesses about the outcome are similar to the hypothesized results, the effect of the experimental intervention may be confounded by demand characteristics (Lush, 2020).

4.1. Hypotheses and preregistration

We hypothesized that, in the instruction-based replication, participants would expect a stronger mean reported experience of the virtual body being theirs when the virtual body moves in synchrony with their real body (in the Synchronous condition), as compared to the Asynchronous condition. Further, we hypothesized that the magnitude of the difference in reported body ownership would be greater for participants presented with such an experiment in Within-subject form as compared to a Between-subjects procedure. The study, with its hypotheses and analysis approach, was preregistered with AsPredicted.[6] The data and analysis code have been archived at OSF.[7]

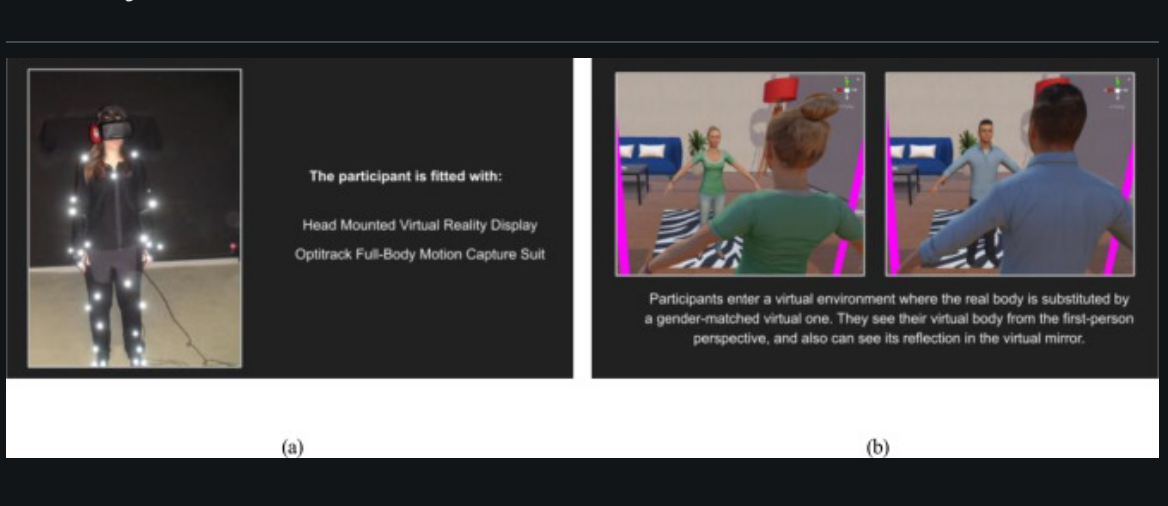
The participant is fitted with:
Head Mounted Virtual Reality Display
Optitrack Full-Body Motion Capture Suit
Participants enter a virtual environment where the real body is substituted by a gender-matched virtual one. They see their virtual body from the first-person perspective, and also can see its reflection in the virtual mirror.
(a)
(b)

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Fig. 3. Screenshots from the video employed to illustrate the study procedure in our instruction-based replication: (a) an outline of the body-tracking and virtual-reality technology setup and (b) an explanation of the VR environment as experienced from the user's perspective.
Banakou and Slater (2014).

4.2. Participants

We recruited 300 people to participate through Prolific. The sample size was chosen as a compromise between statistical power and resource constraints. To ensure both within-subjects and between-subjects studies had comparable power, we doubled the sample size for the between-subjects conditions. All participants were female, to avoid confounding the illusion by way of a gender mismatch between the avatar and participant (the video illustrating the experiment was recorded with a woman). Each participant spent roughly seven minutes on the experiment and was reimbursed with £1 for participating.

Following our preregistered procedure, we removed the data of 94 participants, who correctly guessed the true purpose of our study, or were familiar with body-ownership illusions. The analyses presented below were performed with the remaining 206 participants' data. Note that we found identical effects when examining the complete dataset (see the analysis code at OSF).

4.3. Apparatus

The experiment was implemented via an online survey with a questionnaire hosted on the platform Gorilla.[8] Participants completed the study at their own discretion, using a Web browser on a desktop computer, smartphone, or tablet. For presenting the experiment procedure, we utilized video material accompanying a prior study of body ownership and agency by Banakou and Slater (2014) (see Fig. 3).

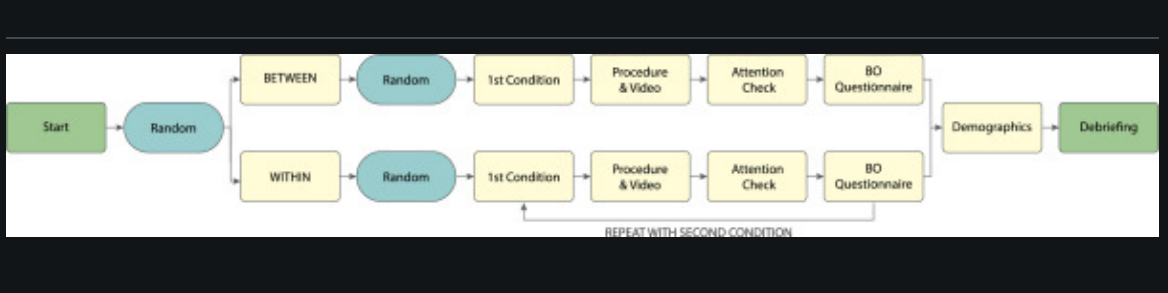

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Fig. 4. Procedure of Experiment II.

4.4. Design

The study employed a between-subjects design with two independent variables: Design and Synchrony.

Design.

The Design variable had two levels: Within-subject and Between-subjects. One-hundred participants were assigned the within-subjects condition whereas 200 were assigned the between-subjects condition, for equal statistical power.

Synchrony.

The independent variable Synchrony had two levels: Synchronous and Asynchronous. Both conditions featured the video and the description of the procedure for the body-ownership illusion in VR. In the Synchronous condition, the virtual body moved along with the tracked real movements of the participant, while in the Asynchronous condition the virtual body moved independently.

4.5. Dependent variables

Body ownership.

Body ownership is the belief that a virtual body experienced through a head mounted display is the participant's own body (Gonzalez-Franco and Peck, 2018). We measured the expectation about the intensity of the body ownership illusion in the thought-experiment context by using a standard questionnaire by Peck and Gonzalez-Franco (2021). The body-ownership score is computed as the mean of six items with a scale of -3 to 3 (for "Strongly disagree" to "Strongly agree").

4.6. Procedure

After an informed-consent and demographic-background stage, we began the experiment proper (the procedure is diagrammed in Fig. 4).

In both conditions, participants were asked to read the description of the body-ownership illusion procedure (see Table 2), then shown video material illustrating the experiment described. English-language onscreen subtitles described the procedure while moving images demonstrated it from first-person perspective. To draw attention to the procedure described in the text and shown in the video, we then asked participants to respond freely to the following question: "What do you think this procedure is supposed to cause? (What is the participant expected to experience?)" After this, we asked them to indicate whether they had heard of the procedure before and to tell us if they had previously participated in an experiment applying this procedure. Participants were then asked to imagine themselves participating in such an experiment and to report the answers they would expect to give, using the Avatar Embodiment Questionnaire (Peck and Gonzalez-Franco, 2021).

In the Within-subject condition, the participant was introduced to both the synchronous and the asynchronous form of the body-ownership illusion in VR, in random order, in keeping with commonly used procedures. In the Between-subjects condition, participants were presented with only one condition (either Synchronous or Asynchronous). This precluded reflecting on the expected outcome in light of the experiment design. They had only the given procedure to refer to.

Table 2. The difference between the two conditions in Experiment II was based on the difference in the procedures described. The table reproduces the descriptions for both conditions verbatim.

	Synchronous	Asynchronous
Description	The participant puts on the virtual reality headset and enters the virtual environment. The virtual environment scene has a room with a mirror. The participant has a virtual body that she can see from first person perspective, and it is reflected in the mirror. The virtual body moves **synchronously** with the real movements of the participant. The participant is asked to do several stretching exercises in order to explore the possibilities of the virtual body.	The participant puts on the virtual reality headset and enters the virtual environment. The virtual environment scene has a room with a mirror. The participant has a virtual body that she can see from first person perspective, and it is reflected in the mirror. The virtual body moves **asynchronously** with the real movements of the participant. The participant is asked to do several stretching exercises, and describe what the virtual body is doing.

4.7. Results

Body ownership.

We compared the mean body-ownership scores between the Asynchronous ($M = 0.03, \quad SD = 1.62$) and the Synchronous ($M = 1.12, \quad SD = 1.20$) condition via mixed-effects two-way ANOVA. The body-ownership score was treated as a dependent variable, condition (Asynchronous, Synchronous) as a predictor, and experiment design (Within-subject, Between-subjects) as an interaction. Descriptive statistics for body-ownership score are presented in Table 3.

The results show that the synchrony; $F(1, 266) = 41.2, \quad p = 6.2^{-10}, \quad$ Cohen's $f = 0.39$, the design of the study; $F(1, 266) = 7.8, \quad p = 0.006, \quad$ Cohen's $f = 0.17$, and their interaction; $F(1, 266) = 12.4, \quad p = 0.0005, \quad$ Cohen's $f = 0.22$ significantly influenced body-ownership scores (as Fig. 5 attests).

To determine if measures in the Synchronous condition are greater than those in the Asynchronous condition, we conducted separate t-tests for body-ownership scores. These tests compared the Synchronous and Asynchronous conditions within both the Between-subject and Within-subject designs. This analysis, which was not preregistered, aimed to assess the adequacy of both within-subject and between-subject experimental procedures to support causal inference.

In the within design, a paired t-test revealed that the Asynchronous condition results in significantly lower body ownership than the Synchronous condition, $p = 5.2 \times 10^{-10}$, Cohen's $d = 0.84$.

For the between design, the difference between Asynchronous and Synchronous conditions was also significant, as an unpaired t-test showed $p = 0.013$, Cohen's $d = 0.42$.

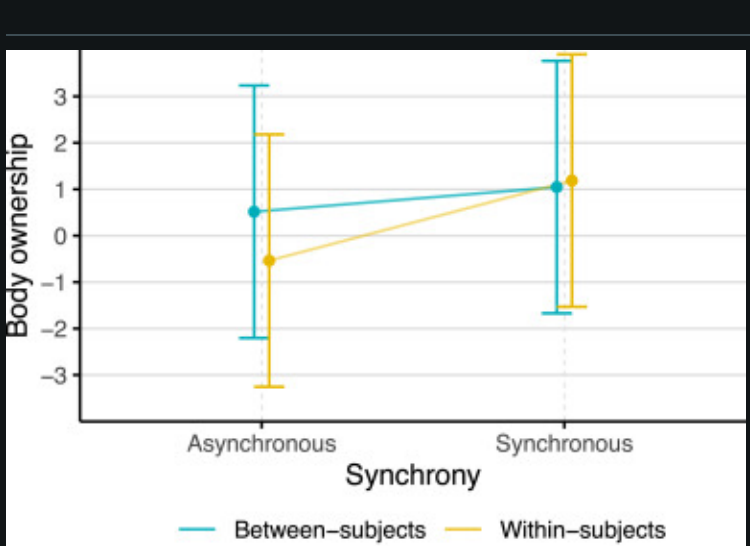

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Fig. 5. Mean reported body-ownership scores. The plot demonstrates an interaction effect between the two independent variables, Synchrony and Design, wherein the difference in body-ownership scores between the synchronous and asynchronous settings was larger with the within-subject design than the between-subjects one. The error bars denote 95% confidence intervals.

Table 3. Descriptive statistics for body ownership scores for condition in Experiment II.

	Between-subject	Within-subject
Asynchronous	$M = 0.52, \quad SD = 1.31$	$M = -0.54, \quad SD = 1.77$
Synchronous	$M = 1.05, \quad SD = 1.22$	$M = 1.19, \quad SD = 1.19$
t-test	$p = 0.013$, Cohen's $d = 0.42$	$p = 5.2 \times 10^{-10}$, Cohen's $d = 0.84$

4.8. Central findings

Even without experiencing the body-ownership illusion themselves, participants were able to predict responses consistent with the effects predicted by embodiment theory and with the findings reported in the literature on these illusions (e.g., Banakou et al., 2013, Peck et al., 2013, Tajadura-Jiménez et al., 2017, Banakou et al., 2016, Mottelson et al., 2023). Clearly, the hypothesized greater experience of body ownership in the synchronous condition could be inferred from the study protocol. The study thus demonstrates demand characteristics exerting an influence through the interaction between the study design and conditions. Namely, participants inferred the researchers' hypothesis solely from the nature of the two conditions presented. It is evident that when the virtual body moves independently of participant movements, scores for constructs that include items such as "I felt like I could control the virtual body as if it was my own body" will be lower than when the participant has full control of the virtual avatar (i.e., than in the Synchronous condition). However, in a within-subject design, the effect is amplified by the ability to compare the conditions with one another. When participants are exposed to two noticeably different conditions, it is obvious that the researcher expects differing responses.

The fact that participants could produce theory-consistent predictions even without experiencing the illusion is testament to how unhindered an experiment's design and protocol can convey the theory that the researcher is testing, even to participants without special knowledge.

5. Discussion

The experiments provide evidence of how participants' susceptibility to demand characteristics transfers to subjective and objective measurements. Data collected through studies with human participants are at substantial risk of confounding. Our experiments taken together attest to this issue's relevance both for artifact evaluations and for more delicate experiential experiments assumed to be free from confounds (Wilson and Soranzo, 2015, Pan and Hamilton, 2018, Parsons, 2015), such as experimental measurements of bodily awareness. They underline the need to exercise caution in interpreting the results of studies in which the research hypothesis can be inferred from the study design. Therefore, our findings are of great importance for the methodological robustness of the empirical research. In the following subsections, we discuss the details of our experiments; consider what is unique about demand characteristics in the context of HCI; and provide recommendations on how to minimize their influence.

5.1. Mechanisms

The underlying mechanisms of demand characteristics' effects are not simple, non direct; they encompass many potential activators. Although we tried to cover the breadth of their effects, some outstanding points remain.

In Experiment I, demand characteristics' effects produced a roughly 3% improvement in performance. This effect, though statistically significant, might appear relatively minor. Even though it is the case in our experiment, we do not assert that demand characteristics in similar experimental designs entirely responsible for the results of the experiment. But they may influence it to such an extent that a seemingly minor effect crosses the significance threshold. Also, typing speed is a challenging metric to manipulate intentionally. The discovery of a significant effect in our study demonstrates that demand characteristics exert *subconscious* effects, which are important in their own right. In studies where participants have more control over their performance (e.g., reaction time or decision-making studies), the effects of demand characteristics may be larger and the confounding may prove much more pivotal.

We did not measure the expectations in Experiment 1, and the improvement in performance can partially be attributed to placebo effects (Boot et al., 2013). The mechanisms are intertwined, and delineating the differences between them appears less feasible than highlighting their confounding nature and emphasizing the necessity of controlling for them.

As for Experiment II, we interpreted the within-subject design's larger-magnitude difference between conditions as arising from the design, making the experimental manipulation obvious. Similarly, Wolf et al. (2022) suggested that research outcomes relying solely on standardized methods, without incorporating a variety of methodological approaches, may be phenomena dependent on those methods alone. For example, introducing asynchrony in body ownership studies might lead participants to focus on the illusion and its intuitively obvious effects on the body ownership scores. Conversely, when the connection between variables is less clear and not obvious from common sense, participants may find it difficult to guess the research hypothesis without explicit cues from the experimenter.

An alternative explanation is that a within-subject design renders it easier to assess the strength of the illusion because the participant has a point of comparison for the experience. We could offer an analogy from wine-tasting. A single glass can let someone appreciate and experience the flavor. However, as multiple samples are tasted, it grows easier to discern the tartness or fruitiness of each wine by comparison. Accordingly, the flavors are pinpointed more accurately and the experience may be more intense. A counterargument might be that if the wines are all reds dominated by the same grape variety, they cannot stand as solid reference points for each other. People in such a setting might not discern the true reason for any distinctions and might rather arbitrarily attribute the differences in flavor to variables such as region or age. However, the question here is what the experiment aims to capture: a pure experience or the nuances of the experience refined through comparison with an alternative. If the objective is to gather subjective estimations of the experience, then between-subject designs might still be a more appropriate choice.

Our studies used crowdsourcing, with an absence of physically present experimenters. Demand characteristics create a bigger problem in lab-based studies because of the social pressure from researchers' presence (Ryan et al., 2013, Joinson et al., 2007), not to mention the part the experimenter plays as a conscious human who can unintentionally provide cues about hypotheses (Doyen et al., 2012). It is important to recognize that the absence of an experimenter does not eliminate the influence of demand characteristics, as demonstrated in our experiments, nor does it shield studies from other experimenter effects. Furthermore, the outcomes and conclusions of research can be affected by experimenter bias during the design and analysis phases, including confirmation bias (Darley and Gross, 1983), leading questions (Schwarz, 1994), specific wordings (Hildum and Brown, 1956), or data manipulation (Holman et al., 2015).

Compensation for participation in experiments can also moderate the behavior and performance of participants in experiments (Corneille and Lush, 2023). Corneille and Lush (2023) suggested that participants in scientific experiments are often motivated by gifts, monetary rewards, or course credits. This creates reciprocal relationships with the experimenter, as well a personal interest in that providing "poor-quality data" might result in rejection of their participation and, therefore, bring them no money or credit points (Corneille and Lush, 2023, Orne, 1962). Hence, participants perform at their best, and misconceptions about what is best might include "I should act in line with the hypothesis", whatever that is perceived to be. How this mechanism operates in the context of crowdsourced research, where participants are paid by default and are unlikely to perceive monetary rewards as an additional incentive to deliver good results, remains unexplored and future research on the mechanisms of demand characteristics' effects in crowdsourced settings is needed.

It is impossible to eliminate demand characteristics entirely. Even when the mechanisms of the effects of demand characteristics are accounted for, experimental participants are still in an experimental situation that is different from their routine and are trying to derive meaning from that situation and, hence, build assumptions. However inaccurate those assumptions may be, they still can taint the study. Nonetheless, focusing the attention of participants away from the primary variable can enhance the validity of the inferences drawn about it.

5.2. Recommendations for scholarship

It is part of human nature to engage in sense-making processes, involving categorization, assumptions, and conclusions. Even in the realm of psychological research, where participation in experiments is subject to these cognitive tendencies, efforts have been made over the past 60 years to acknowledge and address demand characteristics. It is noteworthy that the branch of psychology dedicated to human–computer interaction and the integration of computerized methods in psychological research has, at times, diverted scholarly attention. This diversion may have inadvertently led researchers to overlook or neglect established recommendations for minimizing demand characteristics' effects. Recognizing the significance of these issues, the following paragraph outlines key practices that can enhance experimental research within HCI.

Between-subjects settings

As we demonstrated in Experiment II, between-subjects designs are a good way to reduce the effects of demand characteristics. In within-subject studies, the hypothesis is easier to infer, on account of the difference between experimental conditions. To collect data comparable to previous literature, researchers may still choose to adhere to experimental traditions, or strive to cut costs and effort by opting for within-subject designs, even when the hypothesis enables between-subjects research. Computerized studies have made planning, designing, carrying out, and collecting data much easier, so reduced costs are often not an issue. Needless to say, some research scenarios require within-subject designs — for example, the studies of learning effects or studies with repeated measures, in which observing the same participant is crucial. Apart from such cases, between-subject designs should be favored, as validity is one of the fundamental values of the scientific field.

The recommendation to employ between-subjects designs is not universal. Within-subject designs exhibit higher statistical efficiency, because they minimize subject variance arising from the error terms used for treatment effects (Greenwald, 1976). They also can promote more valid responses in subjective measurements, since anchoring might help the individual make sense of personal experiences in relation to another experience. Also, some experiments are simply not amenable to between-subjects designs, particularly when effects over time are involved. In such cases, the within-subject design should incorporate additional control methods, for validity. Among the options are counterbalancing the order of the conditions, camouflaging the study's purpose, and separating the conditions in time so as to reduce carry-over effects (Greenwald, 1976). We discuss some of these methods in more detail further along.

Double-masking

The literature on demand characteristics is largely confined to documenting their effects in the presence of an experimenter, with some studies singling out the experimenter as the main influence on participant behavior (e.g., Doyen et al., 2012). Double-masking – wherein both participants and experimenters do not know the research hypothesis – minimizes the experimenter's influence on participants' behavior, but, as our experiments illustrate, does not remove them altogether. While in some situations (for example, the development of new interfaces) it may not be possible to achieve complete double-masking, researchers can aim for partial masking. This entails at least one party (either the experimenter or the participant) remaining unaware of the research hypothesis or, alternatively, the purpose being camouflaged by a different hypothesis. In addition, experimenters can be trained in how to interact with participants in a neutral and consistent manner, with emphasis on the importance of avoiding any (verbal or non-verbal) cues that could leak information. The instructions provided to participants should be standardized and clearly written, for further reducing the potential for unintentional cues or hints from the experimenter. Finally, it may be possible to decrease direct interaction between researcher and participant by means of automated prompts, instructions delivered by the interface, or a crowdsourcing technique.

Deception

Double-masking is not always possible, and some studies that do not involve experimenters at all (e.g., in-the-wild or crowdsourced ones) still can suffer the influence of demand characteristics. Deception has effectively reduced their effects in several such cases (Cook et al., 1970, Corneille and Lush, 2023). One can mask the expectations (for example, by omitting prior information or by presenting a cover story (Weber and Cook, 1972, Halliwell and Dittmar, 2004, Lindsay and Johnson, 1989, Westermann et al., 1996)), or distract participants (through the so-called red herring technique (Laney et al., 2008)). Psychology research points to possible benefits in using a "Swiss cheese model" (Wexley and Baldwin, 1986, Olson and Raz, 2021), wherein the design implements several levels of deception.

It should be noted, that deception involves ethical considerations, that from the perspectives of some fields might be viewed as unacceptable (Brannigan, 2004). If deception applies, debriefing of participants is a necessary final step of the study.

Indirect measurements

A common recommendation for mitigating the effects of demand characteristics is to use indirect (or implicit/proxy) metrics with which participants are not in full control of their responses (e.g., the Implicit Association Test (Greenwald et al., 1998)). However, this method, too, should be treated with caution. After all, Experiment I showed that demand characteristics can confound even objective measurements that one might presume to reside outside participants' control. Furthermore, suggestion effects have been documented in settings with indirect measurements (Lush et al., 2020).

Post-experiment interviews and discussion

When the influence of demand characteristics cannot be prevented, one possible strategy for controlling them is to use post-experiment interviews and questionnaires, in which participants report their expectations of the study and reflect on their behavior in the experiment. A common technique is to ask participants an open question about what they think the study was about or conduct debriefing questionnaires (e.g., Bargh and Chartrand, 2000). Alternatively, one can employ the Perceived Awareness of the Research Hypothesis Scale (Rubin, 2017). In light of post-experiment information, the protocol can filter out participants who guessed the hypothesis correctly, or their scores on the scale can be considered in the analysis as a covariate (Rubin, 2017). It is worth noting, though, that post-experimental inquiries might themselves be subject to demand characteristics' effects. Participants could guess that the post-experimental interview is intended to identify whether they guessed the hypothesis correctly and suspect that the researcher wants them to be unaware of the true purpose of the study (Nichols and Maner, 2008). While this problem is known (Corneille and Béna, 2023, Corneille and Lush, 2023), there are no universal and agreed-upon techniques for mitigating the presence of demand characteristics in the post-experimental inquiries. One way of avoiding them is to ask participants for honesty in responses and highlight that this honesty will ensure the validity of the experimental results. Another way of overcoming the limitations of post-experimental questions is the use of quasi-controls or instruction-based replications (Corneille and Lush, 2023, Orne, 1969).

Pre-experiment inquiries

Pre-experiment controlling for confounds is a relatively uncommon method but might be worth considering. It usually takes the form of thought experiments, or instruction-based replications, in which participants are asked to predict the effects of manipulation or to try to behave as if they were in a treatment group. If predicted responses are consistent with the expected outcomes of the planned experiments, then the experimental procedure might communicate demand characteristics to participants, or simply be common sense knowledge (Corneille and Béna, 2023). As an example of using instruction-based replication to identify the presence of demand characteristics in the experimental situation, in the rubber-hand illusion study by Lush et al. (2020), most participants correctly guessed the effect of the illusion after merely being presented with the experiment design. Conducting a pre-experiment inquiry stage could equip researchers to modify the design to minimize the confounding effects of demand characteristics.

5.3. The subtlety of demand characteristics in human–computer experiments

Studies of computer use and human–computer interaction are often considered immune to demand characteristics, as task instructions and data collection is automized. Nevertheless, the tasks and behaviors studied experimentally are often quite complex and centered around the use of technology. Thus, in many cases, researchers studying computer use and interaction in experimental research do not want to isolate the factors from the usage situations. For example, when working with user interfaces it is hard and often not possible to mirror effective controls from experimental psychology. For instance, non-treatment conditions are hard to develop in studies involving evaluation of technology. In classical psychological experiments, researchers can leave a control group of participants unmedicated, or unprimed and observe their behavior, without exposure to the intervention. When evaluating the effectiveness of a new artifact, a baseline against the experimental developments are needed. Thus, in both conditions, participants inevitably have to interact with technology, which makes the difference between the two conditions visible. The implication is, that studies of computer use and interaction, contrary to belief, may intensify the effect of demand characteristics.

The effects of demand characteristics stand out from other biases in empirical research because, unlike statistical mistakes or questionable research practices, they are subtle (Orne, 1962). Neither researchers nor participants might recognize the demand characteristics bound up with an experimental situation (Corneille and Lush, 2023), in general and in specific cases. This may be due, in part, to a lack of awareness that such confounding factors exist or the assumption that the study design has cleansed the experiment of such influences. The confounds might be also perceived as benign, because the standardized and validated scales and objective measurements appear reassuringly safe. The very term "objective" implies being free of subjective influence and suggests that measurements assigned this label are immune to confounding factors. We showed that this is not the case, even in studies following common designs.

While placebo effects and demand characteristics in experimental studies of computer interaction have gained in prominence over the years (e.g., Popp and Schneider, 2015, Linde et al., 1997, Baskin et al., 2003), the attention is much less overt compared to other strands of empirical research that employs human participants. Given the inherent focus on the interactions between humans and technology in studies on human–computer interaction, the scrutiny of potential sources of bias and confounding variables should receive sufficient attention.

5.4. Implications

Theory and knowledge

In the last decade, it has become apparent that many fields of science are suffering from a replication crisis, with computer science being no exception (Cockburn et al., 2018, Wacharamanotham et al., 2020, Greenberg and Thimbleby, 1992, Cockburn et al., 2020). Multiple attempts to reproduce the results of prior scientific studies have come up empty-handed, clarifying that most empirical results are tricky to replicate, at best. The validity of empirical knowledge in many sciences has come into question. Awareness of demand characteristics and their confounding influence on the experiments contribute to the uncertainty about validity further. How much of what we "know" from experiments is an artifact of suggestion effects? It would be tempting to ignore this issue, but we do have a way forward.

The replication crisis has brought about new experimental standards, calling for, among other things, open-science practices, replication attempts, and larger sample sizes (Shrout and Rodgers, 2018). While controlling for demand characteristics is seldom mentioned in guidelines for good research practices, it is clear that reducing their presence can enhance the validity of studies of computer use and interaction, as well as promote empirical robustness, and contribute to the evolving standards for human–computer experiments, alongside practices of preregistration, transparent reporting, data-sharing, and communicating statistical uncertainty (Dragicevic et al., 2019, Wacharamanotham et al., 2023).

We encourage scholars to do more replication work, with an eye to reducing the presence of demand characteristics. To this end, the recommendations outlined above provide a starting point for refining and strengthening both empirical knowledge and theory.

Human–computer interaction

While we have addressed the impact of demand characteristics on human–computer experimentation, such methods of empirical inquiry as surveys, interviews, and field trials experience similar confounding, related to the expectations/beliefs of participants and researchers alike. The literature documents many instances of studies confounded by human beliefs and responses to contextual factors. Participants experience placebo effects when interacting with new technologies (Kosch et al., 2023, Denisova and Cairns, 2015, Finniss et al., 2010), adjust their evaluations of artifacts to the preferences of the investigator (Dell et al., 2012), change their use of the technology to give the researcher "good" data (Brown et al., 2011), and express socially desirable opinions and judgements about the content (Vashistha et al., 2018).

Humans are aware and social. We situate our actions in context, read cues, and interpret what is happening to us. Therefore, being sensitive to said fundamental aspect of human behavior is imperative. This profound realization extends beyond any single type of study and stretches across discipline boundaries. For empirical research in the field of human–computer interaction in particular, it demands a new benchmark for taking human consciousness into consideration, both in the study-design phase and when one derives conclusions from the empirical findings.

6. Conclusion

Participants read cues of numerous sorts from the situations they encounter in experiments, not least about the hypothesis under testing. Effects such as adjusting one's behavior and other responses to match pose a serious threat to experiments' validity. What we think we have found in experiments with human participants might not be a result of the hypothesized mechanism, but a suggestion effect. Our two experiments attesting to the potency of demand characteristics are especially important to human–computer experiments since they illuminate the vulnerability of both subjective and objective measurements. Accordingly, we call for studies of computer use and interaction to put greater stress on controlling for demand characteristics. The recommendations we offer may aid in mitigating their effects.

CRediT authorship contribution statement

Olga Iarygina: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Kasper Hornbæk:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Aske Mottelson:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Olga Iarygina reports financial support was provided by Pioneer Centre for Artificial Intelligence. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

I have shared the link to my data and code in the manuscript.

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View in Scopus ↗ Google Scholar ↗

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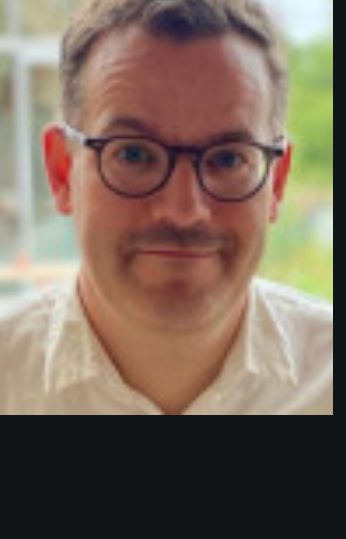

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